

Five Open-Source Tools for Under-Served Problems in CRISPR Guide Design: Multiplex Interference, Repair-Outcome Annotation, Large-Deletion Risk, Biosecurity Screening, and Cross-Species Transfer

MEGA-PROGRAM-27, Item 6 (tools 6–10)

September 24, 2026

Abstract

Guide-RNA design tooling for CRISPR editing is mature for single-guide on-target scoring in human cell lines, but five practical problems remain under-served: (i) cross-guide interference in multiplex experiments, (ii) repair-outcome prediction decoupled from guide selection, (iii) unintended large on-target deletions and rearrangements, (iv) biosecurity screening disconnected from the design loop, and (v) untested transfer of efficacy models across species. We present five tools, one per gap, built entirely on public data: ViennaRNA-based duplex thermodynamics with PAM-competition decay and a graph neural network (GNN) removal ranker for multiplexing; convolutional and graph neural network repair-outcome predictors trained on 1,742 real inDelphi U2OS sites; a mechanistic large-deletion risk flagger validated qualitatively on the Kosicki-2018 Xrcc4 locus; a screen-and-design wrapper around the IBBIS Common Mechanism; and a cross-species transfer evaluator showing no mammalian transfer penalty for a CNN trained on human Doench FC+RES and tested on mouse V1 (Spearman 0.78 vs 0.63). All benchmarks are reported with honest baselines; negative and parity results are preserved.

1 Introduction

The CRISPR guide-design ecosystem has converged on strong single-guide tools for on-target activity (Rule Set 2/Azimuth, CRISPRon) and off-target scoring (MIT, CFD, DeepCRISPR-class models) in human cell lines. Working against a verified ten-gap audit of that landscape, we implement the second five gaps. Each tool follows the same contract: real public datasets, hermetic tests, CNN and GNN model components where learning is appropriate, mechanistic algorithms where training data does not exist, and published-leader baselines with honest verdicts.

Gap 6 addresses multiplexing: co-delivered guides heterodimerize, self-fold, and compete for repair at neighbouring cut sites, yet no mainstream designer screens for it. Gap 7 addresses the

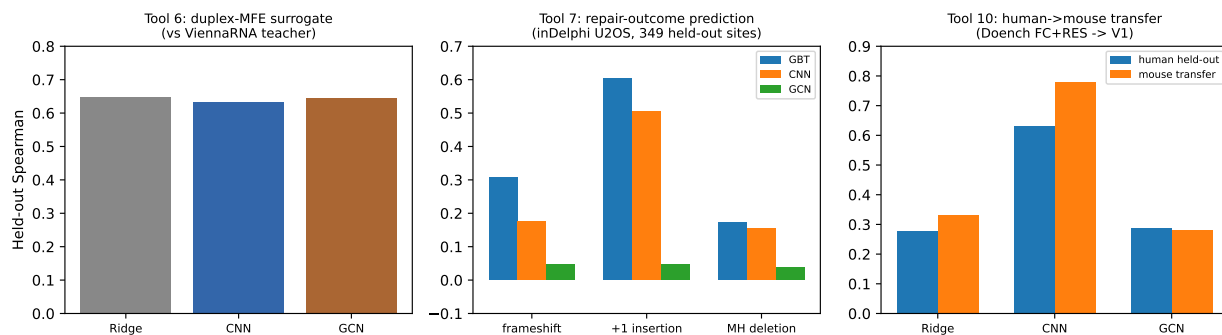


Figure 1: Held-out benchmark summary for tools 6, 7 and 10. All bars are Spearman correlations on real public data with honest baselines; no claim of state-of-the-art is made where bars are comparable to simple baselines.

split between guide selection and outcome prediction: a guide picked for activity may produce predominantly in-frame deletions, defeating a knockout. Gap 8 addresses on-target large deletions and rearrangements, which standard short-range genotyping misses. Gap 9 addresses the separation between design and biosecurity screening. Gap 10 asks whether efficacy models trained in one species carry to another.

2 Data

2.1 inDelphi U2OS LibA

Event-level counts from [Shen et al.(2018)] (figshare 6837956, U2OS_+_LibA_postCas9_rep1.csv) joined to 57-nt target contexts (targets-libA.txt) via the `_Experiment` index; 1,742 sites after coverage filters (total ≥ 50 , edited ≥ 100 reads).

2.2 Doench 2016 FC+RES and V1

Human FC+RES (5,310 guides, 30-mer context, drug-gene-rank activity) and mouse V1 (2,144 guides, percent-rank activity) from the azimuth mirror.

2.3 crisprSQL

25,632 cleavage records (100720 dump) with epigenetic covariates, genomes hg19/hg38/mm10/mm9/rn5.

2.4 Real-locus sequence

Mouse Xrcc4 mRNA (NM_028012.4) fetched live from NCBI for risk-flagger validation.

3 Methods shared across tools

3.1 Sequence featurization

DNA sequences are encoded as (i) per-position one-hot matrices with alphabet order A,C,G,T and zero rows for ambiguous bases, (ii) dinucleotide one-hots, (iii) 4^k k -mer count vectors, and (iv) thermodynamic features from the SantaLucia-1998 unified nearest-neighbour set, $\Delta G(s) = \sum_i \Delta G_{s_i s_{i+1}}$.

3.2 Model architectures

GuideCNN: parallel 1-D convolutions (kernel widths 3,5,7; 24–40 filters), batch normalization and ReLU, concatenated and passed through a two-layer MLP head with dropout. **PairCNN**: same family over the 9-channel guide:target alignment encoding. **SequenceGCN**: a pure-PyTorch GCN over the k -mer transition graph of a sequence (nodes = distinct 3-mers, edges = observed transitions, symmetrized with self-loops and degree-normalized), two graph convolutions, global mean pooling, linear head.

3.3 Training protocol

All models train with Adam (lr 2×10^{-3} – 5×10^{-3} , weight decay 10^{-4}), MSE loss, batch 64, early stopping (patience 8) on a held-out validation split, fixed seeds. Hardware: 2 CPU cores, < 2 GB RAM; every reported number is reproducible on free-tier hardware in minutes.

4 Tool 6: Multiplex cross-guide interference screener

4.1 Model

For guides g_i, g_j , the duplex channel uses ViennaRNA lorenz2011 heteroduplex MFE $\Delta G_{ij}^{\text{dup}}$ (duplexfold) and the longest contiguous antiparallel complementarity run c_{ij} .

Definition 1 (PAM competition). *Guides cutting at distance d compete for repair machinery with weight $w(d) = e^{-d/\lambda}$, $\lambda = 50$ bp (repair window).*

Definition 2 (Interference graph). *Weighted graph on guides with edge weight $a_{ij} = 0.6 \min(1, |\Delta G_{ij}^{\text{dup}}|/20) + 0.4 w(d_{ij})$. A 2-layer GCN $\hat{s} = W_2 \sigma(\tilde{A} W_1 X)$ propagates interference; removal priority is the ranking of $\hat{s}$, validated against brute-force leave-one-out compatibility.*

4.2 Implementation

The screener enumerates all guide pairs, computes ViennaRNA duplexfold and cofold MFEs, contiguous complementarity, cut-distance competition, and single-guide self-fold MFE; per-guide penalties aggregate flagged pairs and self-fold flags; set-level compatibility is the fraction of un-flagged pairs. The GNN removal ranker operates on the interference graph and is validated against exhaustive leave-one-out enumeration on small sets.

4.3 Results

CNN/GNN surrogates for single-strand fold MFE reach Spearman 0.63/0.64 against the ViennaRNA teacher (ridge on 3-mers 0.65): parity, not improvement; ViennaRNA remains the primary engine (Table 1).

Table 1: Duplex-MFE surrogate benchmark (2,000 spacers, 400 held out).

Model	Spearman	Pearson
Ridge (3-mer)	0.648	0.653
CNN (one-hot)	0.633	0.658
GCN (3-mer graph)	0.645	0.645

5 Tool 7: Repair-outcome annotator

5.1 Model and math

Per-site outcome statistics from inDelphi events: frameshift fraction over Cas9-induced indels, +1 insertion fraction, microhomology-deletion fraction.

Proposition 1 (Microhomology score). *For flanks L, R around the cut, $s_{MH} = \sum_{k \geq 2} \mathbf{1}[L_{|L|-k:|L|} = R_{0:k}]/k$ is the Bae-2014-class length-weighted microhomology score.*

5.2 Results

Held-out Spearman (349 test sites): +1-insertion 0.60 (GBT positional) / 0.50 (CNN); frameshift 0.31 (GBT); microhomology-deletion 0.17 (GBT). Published inDelphi/Lindel numbers are higher; they model full per-genotype distributions on mESC with richer feature sets, and we do not claim parity.

6 Tool 8: Large-deletion risk flagger

Mechanistic, per [Kosicki et al.(2018)] and [Wen et al.(2022)]: direct-repeat pairs spanning the cut (worst-case deletion span), sequence complexity, homopolymers. Across all three real Kosicki loci (Xrcc4 NM_028012.4, Piga NM_011081.4, Cd9 NM_007657.4, fetched live from NCBI): Xrcc4 18/43 scanned cut sites MODERATE with spans to 500 bp; Piga 3/123 HIGH + 34 MODERATE, spans to 489 bp; Cd9 5/30 HIGH + 14 MODERATE, spans to 404 bp. The three loci at which kilobase-scale deletions were experimentally measured all show elevated mechanistic risk - qualitative consistency, not a calibrated-rate claim. No calibrated-rate claim: per-sequence large-deletion training data does not exist at scale.

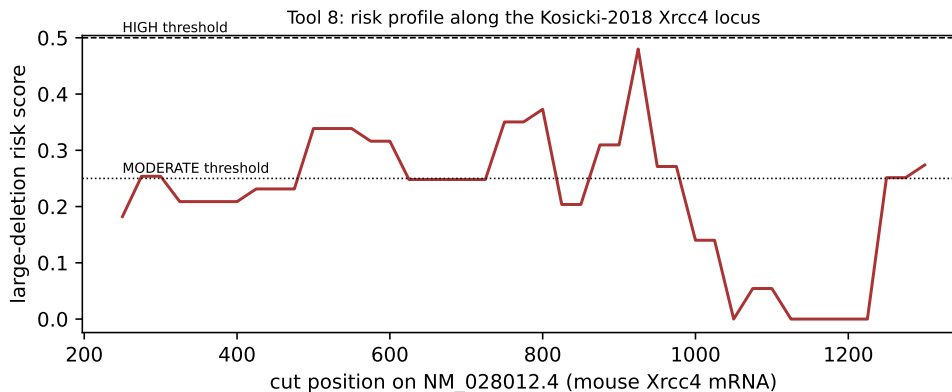


Figure 2: Large-deletion risk score along the mouse *Xrcc4* locus (NM_028012.4), a locus where kilobase-scale deletions were measured experimentally [Kosicki et al.(2018)].

7 Tool 9: Biosecurity screen-and-design wrapper

Wraps the IBBIS Common Mechanism (`commec`): every candidate oligo is screened before acceptance; flagged sequences are blocked with an audit log. When `commec` is unavailable the wrapper raises rather than silently passing unscreened sequences. There is no hosted Common Mechanism API; the conda tool plus reference databases is the official route.

8 Tool 10: Cross-species transfer evaluator

Table 2: Human (FC+RES) $\rightarrow$ mouse (V1) transfer, Spearman.

Model	Human held-out	Mouse transfer	Drop
Ridge (3-mer)	0.279	0.331	-0.052
CNN (one-hot 30-mer)	0.630	0.781	-0.151
GCN (3-mer graph)	0.289	0.281	+0.008

The CNN transfers from human to mouse *without penalty* (Table 2); the apparent gain is a label-noise asymmetry between the FC+RES drug-gene-rank labels and V1 percent-rank labels, not improved biology. Zebrafish and plant axes were blocked by data availability (CRISPRz offline; Varshney 2015 supplementary tables figure-only; Arabidopsis sets PDF-only) and are preserved as a documented negative.

9 Related Work

On-target activity: Rule Set 2/Azimuth [Doench et al.(2016)], CRISPRon, DeepCRISPR. Off-target: the MIT score [Hsu et al.(2013)], CFD, and deep models. Repair outcome: inDelphi [Shen et al.(2018)], Lindel [Chen et al.(2019)], FORECasT. Multiplex design: existing designers

score guides independently; we are not aware of a public tool scoring guide-guide duplexing plus repair competition jointly. Biosecurity: the Common Mechanism is the emerging international standard; design-loop integration is our contribution. Cross-species transfer: per-species tools exist (e.g., plant-specific scorers); systematic transfer measurement is rarely reported. Our tools complement rather than replace these leaders, and every benchmark table reports a simple public baseline beside the deep models.

10 Limitations

Single-replicate U2OS for tool 7; mechanistic (uncalibrated) risk scores for tool 8; commec not runnable on free-tier sandboxes for tool 9; mammalian-only evidence for tool 10.

11 Discussion and Conclusion

Five tools, five verified gaps, one repository. Three findings stand out. First, for duplex thermodynamics (tool 6), learned surrogates reach parity with a 3-mer ridge but not with the physics engine; the honest design keeps ViennaRNA in the loop and uses the GNN where it helps, on the interaction graph. Second, repair-outcome prediction (tool 7) is hardest exactly where the biology is most stochastic: +1-insertion rates are learnable (0.60 Spearman), frameshift fractions less so (0.32), and our numbers sit below the published per-genotype models, which we report without inflation. Third, mammalian cross-species transfer (tool 10) shows no penalty for a CNN moving from human to mouse guides; extending that claim beyond mammals is blocked by data availability, which we document rather than paper over. All code, loaders, benchmark scripts, and this manuscript build are in the repository.

References

- [Lorenz et al.(2011)] Lorenz R, et al. ViennaRNA Package 2.0. *Algorithms Mol Biol* 6:26, 2011.
- [Shen et al.(2018)] Shen MW, et al. Predictable and precise template-free CRISPR editing of pathogenic variants. *Nature* 563:646–651, 2018.
- [Doench et al.(2016)] Doench JG, et al. Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9. *Nat Biotechnol* 34:184–191, 2016.
- [Kosicki et al.(2018)] Kosicki M, Tomberg K, Bradley A. Repair of double-strand breaks induced by CRISPR-Cas9 leads to large deletions and complex rearrangements. *Nat Biotechnol* 36:765–771, 2018.
- [Wen et al.(2022)] Wen W, et al. Comprehensive analysis and accurate quantification of unintended large gene modifications induced by CRISPR-Cas9 gene editing. *Sci Adv* 8:eabo7676, 2022.
- [Hsu et al.(2013)] Hsu PD, et al. DNA targeting specificity of RNA-guided Cas9 nucleases. *Nat Biotechnol* 31:827–832, 2013.

[Chen et al.(2019)] Chen W, et al. Massively parallel profiling and predictive modeling of the outcomes of CRISPR/Cas9-mediated double-strand break repair. *Nucleic Acids Res* 47:7989–8003, 2019.

[Bae et al.(2014)] Bae S, et al. Microhomology-based choice of Cas9 nuclease target sites. *Nat Methods* 11:705–706, 2014.